

Lab 3: Sampler and DTFT

Instructions:

1. Lab report to be submitted (on Nalanda) at the end of the lab hours (preferably), late submission will result in loss of marks, use the lab hours to finish the work
2. All the tasks in each lab have to be in one main MATLAB file with “.m” extension, differentiate each question by commenting. Alternatively, you can zip all .m files together, name them appropriately and upload one zip/rar file. Make one Word/pdf file with all the results corresponding to the questions
3. Plots should include x label, y label, legend, and titles
4. File name: ID_Lab3.m/ID_Lab3.rar or zip **and** ID_Lab3.pdf
5. If the report/code is found to be copied, zero marks will be awarded **[Ensure that you show the outputs of the task to the TAs and get them evaluated. No queries will be entertained post lab session.]**

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1. The operation of signal dilation (or decimation or down-sampling) is defined by $y(n) = x(nM)$ in which the sequence $x(n)$ is down-sampled by an integer factor M , as discussed in the lecture.

- a) Develop a MATLAB function `dnsample` that has the form

```
-----  
function [y,m] = dnsample(x,n,M)  
% Downsample sequence x(n) by a factor M to obtain y(m)  
-----
```

to implement the above operation. Use the indexing mechanism of MATLAB with careful attention to the origin of the time axis $n = 0$.

- b) Generate $x(n) = \sin(0.12c\pi n)$, $-50 \leq n \leq 50$. Decimate $x(n)$ by a factor of d to generate $y(n)$. Plot $x(n)$ and $y(n)$ using a subplot and comment on the results. Here, c and d are the last and second last digits of your ID. If any digit is zero, add 7.

DTFT: There is no direct command in MATLAB to implement the DTFT. The magnitude and phase need to be implemented separately using the formula, or one can follow the matrix approach as given below, or you can come up with any other method.

If $x(n)$ is of finite duration, then MATLAB can be used to compute $X(e^{j\omega})$ numerically at any frequency ω . The approach is to implement directly. If, in addition, we evaluate $X(e^{j\omega})$ at equispaced frequencies between $[0, \pi]$, then can be implemented as a *matrix-vector multiplication* operation. To understand this, let us assume that the sequence $x(n)$ has N samples between $n_1 \leq n \leq n_N$ (i.e., not necessarily between $[0, N - 1]$) and that we want to evaluate $X(e^{j\omega})$ at

$$\omega_k \triangleq \frac{\pi}{M}k, \quad k = 0, 1, \dots, M$$

which are $(M + 1)$ equispaced frequencies between $[0, \pi]$. Then

$$X(e^{j\omega_k}) = \sum_{\ell=1}^N e^{-j(\pi/M)kn_{\ell}} x(n_{\ell}), \quad k = 0, 1, \dots, M$$

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When $\{x(n_\ell)\}$ and $\{X(e^{j\omega_k})\}$ are arranged as *column* vectors $\mathbf{x}$ and $\mathbf{X}$, respectively, we have

$$\mathbf{X} = \mathbf{W}\mathbf{x}$$

where $\mathbf{W}$ is an $(M+1) \times N$ matrix given by

$$\mathbf{W} \triangleq \left\{ e^{-j(\pi/M)kn_\ell}; n_1 \leq n \leq n_N, k = 0, 1, \dots, M \right\}$$

In addition, if we arrange $\{k\}$ and $\{n_\ell\}$ as *row* vectors $\mathbf{k}$ and $\mathbf{n}$ respectively, then

$$\mathbf{W} = \left[\exp \left(-j \frac{\pi}{M} \mathbf{k}^T \mathbf{n} \right) \right]$$

In MATLAB we represent sequences and indices as row vectors; therefore taking the transpose we obtain

$$\mathbf{X}^T = \mathbf{x}^T \left[\exp \left(-j \frac{\pi}{M} \mathbf{n}^T \mathbf{k} \right) \right]$$

Note that $\mathbf{n}^T \mathbf{k}$ is an $N \times (M+1)$ matrix. Now can be implemented in MATLAB as follows.

```
>> k = [0:M]; n = [n1:n2];  
>> X = x * (exp(-j*pi/M)) .^ (n'*k);
```

2. Using the matrix-vector multiplication approach discussed above (or any other approach you can come up with), write a MATLAB function to compute the DTFT of a finite-duration sequence. The format of the function should be

```
-----  
function [X] = dtft(x,n,w)  
% Computes Discrete-time Fourier Transform  
% [X] = dtft(x,n,w)  
% X = DTFT values computed at w frequencies  
% x = finite duration sequence over n  
% n = sample position vector  
% w = frequency location vector  
-----
```

Use this function to compute the DTFT $X(e^{j\omega})$ of the following finite-duration sequence over $-\pi \leq \omega \leq \pi$. Plot DTFT magnitude and phase graphs in one figure window.

$$x(n) = (0.6)^{|n|} [u(n+a) - u(n-b)].$$

Comment on the phase plot obtained.

Here, a and b are the last and second last digits of your ID. If any digit is zero, add 12.
